

Regime-Aware LLM Agents for Cryptocurrency Markets

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Abstract—Large Language Models have been applied to financial markets successfully in recent years and have shown remarkable capabilities regarding decision making and portfolio management. However, these LLM-based trading agents are mostly based on a static approach which makes them one-size-fits-all type. These do not take into account the dynamic nature of the market and different market regimes. Especially, cryptocurrency markets and prices are characterized by extreme volatility, regime shifts and co-dependency among the coins. Static prompts are not suitable for them, and result to inadequate performance. This document proposes a Regime-Aware LLM Agent (RALA) which is a novel framework that first detects regimes using Maximum Mean Discrepancy (MMD) and then routes the decision making process through specialized LLM prompts for different regimes and specific market conditions. We have also taken into account diverse market conditions like the FTX collapse and ETF approval events which causes a sudden change in the market regime and also a trend shift. Our results demonstrate that this regime-aware strategy given an improvement of 34% in risk-adjusted returns when compared to the state of the art static prompt LLM model.

Index Terms—Large Language Models, Cryptocurrency Trading, Market Regime Detection, Maximum Mean Discrepancy, Prompt Engineering, Algorithmic Trading.

I. INTRODUCTION

There are a wide range of tradable markets in the world, equity markets of various different countries, cryptocurrency markets, forex, commodities, etc. Out of them the cryptocurrency markets exhibit the most volatility and uncertainty. Bitcoin experiences annualized volatility of 65-90% compared to 20-25% for major equity indices [1], [2]. This volatility is not uniformly distributed along the time scale but is clustered into various different regimes which are short periods of high turbulence and changes. In between these clusters are prolonged periods of market lull or calmness characterized by range bound price action and low volatility. These changes in market regime creates a unique challenge for LLM based automated trading and portfolio management systems [ref]. Cryptocurrency markets are also open 24/7 which leads to an absence of circuit breakers and any global news has an immediate effect on the prices. The prevalence of retail buying and selling in large quantities also amplifies the volatility and gives more importance to regime - dependent dynamics [4].

Recent advancements in the field of Large Language Models have vastly improved the financial reasoning capabilities.

Recent frameworks like FinGPT demonstrate LLMs potential for sentiment analysis, trading alpha generation, and even in direct financial and trading decisions [5]. Some examples like TradingAgents and FLAG-Trader integrate LLMs with reinforcement learning with generally leads to enhanced trading performance in low volatility markets and simulations [6], [7]. These hybrid trading approaches like TradingAgents are based on an hierarchical framework with analyst teams, research people and external risk management agents. FLAG-Trader was the first to work with a gradient-based reinforcement learning model for LLM finetuning in financial contexts. FLAG-Trader demonstrated that parameter-efficient adaptation enhances performance [7], [8]. Both of these approaches focus on agent architecture and LLM feature engineering while keeping the market condition and volatility windows as static backgrounds rather than going for dynamic windows which may require more resources and computation but is expected to give better results and also makes them more real world compatible.

LLMs trading agents are generally categorized into news-driven, reasoning driven, and reinforcement-learning driven architectures [9]. As the name suggests, news driven agents integrate retail sentiment from financial headlines [10], reasoning-driven agents are based on pre-learned trading strategies and pattern recognition and employ reflection and debate mechanism [11]. RL agents use rewards and optimize their working through back testing and Monte-Carlo strategies [12]. However, none of these agents take into consideration market regimes, volatility and trends in any capacity. Next we would learn about market regime detection especially in the case of cryptocurrency high volatility markets.

A. The Problem

The current state of the art work that uses LLM trading agents treats any financial market as one entity instead of breaking it down into different windows each characteristic of a bull market, bear market, sideways consolidation and news-driven price movements. An LLM prompt which is designed for aggressive bull markets and trend following may be prone to large drawdowns in a bear market or in a range bound event driven one. This regime independent approach is a direct contradiction of established finance principles and learning where volatility clustering and regime-aware models

have proven to be essential and hence better performing [13], [14].

In the text that follows we will explain our proposed trading system which works on identifying similar trends in the markets based on previous movements and then using an LLM prompt which is optimized for better returns and lower portfolio drawdowns for that specific market regime.

B. Key Contributions

We propose Regime Aware LLM Agents (RALA) which adapts LLM prompts based upon the detected market regimes. Our framework is based upon maximum mean discrepancy (MMD) - which is a kernel based metric to detect shifts in the distribution of cryptocurrency returns, hence dividing the market regime into high-volatility and low-volatility periods [15], [16]. After regime detection, the trading decisions are fed via specialized prompts and pre-determined rules, conservative risk-off prompts for high volatility periods, aggressive risk-on prompts for stable market regimes, and neutral conditions for in-between transition periods. This can also be extended to specialized periods for news headlines like tariffs, wars, etc.

- MMD based market regime detection. This is based upon non-parametric detection of distribution shifts in market behavior using kernels in the Reproducing Kernel Hilbert Space (RKHS)
- Dynamic Prompting. There is automated selection of regime - optimized prompts for LLM agents available.
- Contextual Risk Management which helps with position sizing and risk constraints.

Our experiments are based on Bitcoin (BTC/USD) hourly data available via Coinbase from January 2022 to December 2024. It demonstrates that our Regime-Aware LLM Agents achieve superior risk - adjusted returns compared to both static LLM agents (RALA) and traditional technical strategies used by a floor trader. During the high - volatility and outlier events like the FTX collapse (November 2022), RALA's conservative and risk-off detection limited the max drawdown across simulations to -12% in comparison to -31% for static agents. Similarly during the 2024 Bitcoin-ETF approval rally, it captured as estimate of 89% of the upside move while maintaining the specified risk profile explained in the experiments section.

II. RELATED WORK

Current state of the art large language model based automated financial systems work using highly optimized prompts which instruct the LLM on how to behave while trading using a strict set of constraints and rules in order to generate profit [9], [10]. These LLM agents are categorized into different types based on how they work and what are they meant for. Any highly volatile market asset like cryptocurrency has different market regimes which have inherent characteristics and would need to be identified for the LLM to be useful. This section is a study about the existing LLM models and market regimes which need to be combined in order to attain significant improvement in trading performance.

A. Large Language Model-Based Financial Trading Agents

The use of LLMs in the field of algorithmic trading has been evolving rapidly and FinGPT is being used for sentiment analysis and robo-advisory applications [5], [9]. These are basically open-source financial LLMs. Further work explored multi-agent architecture where LLMs are used for different tasks like strategists, analysts, and execution agents under various conditions [6].

Furthermore, TradingAgents proposed a hierarchical framework for running a full-scale system with analyst teams, research teams, risk management agents, etc. This demonstrated that collaboration between different LLM agents improves decision quality and better results [7]. FLAG-Trader introduced gradient-based reinforcement learning agents for LLM fine-tuning in trading contexts which lead to better parameter adaptation and lower portfolio drawdown [7], [8]. All of these approaches primarily focus on agent architecture and learning mechanisms while treating market conditions as static backgrounds rather than dynamic contexts hence requiring adaptation.

Recent surveys have categorized LLM trading agents into news-driven, reasoning-driven and RL-driven architectures [9]. News-driven agents integrate sentiment from financial headlines like an exchange collapse, war news, peace talks and FED interest rate decisions [10]; reasoning-driven agents employ reflection and rebate mechanism which is often slow [11]; RL based agents which use back-testing and rewards if moves in the intended direction. Being said that non of these agents and methods explicitly model market regime transitions or adapt prompts based on volatility conditions and they often use one-approach fits all as their method which is mostly not the case as the financial markets are dynamic in nature and are inherent to change. Next we would talk about market regime change in cryptocurrency as a method to employ different tactics in different market conditions based on their volatility.

B. Market Regime Detection in Cryptocurrency

Cryptocurrency markets exhibit much higher volatility than traditional markets and also a pronounced regime-dependent behaviors which makes it distinct from traditional assets like stocks, gold, silver and crude oil. A characteristic known as volatility clustering is significantly stronger in the case of cryptocurrency and digital assets due to 24/7 trading, lower liquidity and a retail-driven sentiment [1]. In volatility clustering, high/low volatility periods for the given asset are clustered together respectively and can be considered as a mini-window of similar behaviors and characteristics. Generalized Autoregressive Conditional Heteroskedasticity (GARCH) family models and regime-switching approaches have been adapted for cryptocurrency volatility forecasting [13], [14]. GARCH models time-variance by assuming that it depends on past squared errors and part price variance, hence capturing volatility clustering in a way. In our work we have used maximum mean discrepancy(MMD) as a tool for similarity detection [15]. MMD is non-parametric and an exact metric which can be used for detecting distribution shifts in financial

time series. Unlike Markov-switching models that assumes specific transition dynamics, MMD detects regime changes by measuring the distance between two return probability distributions in a Reproducing Kernel Hilbert Space(RKHS) in high dimensions [15], [16]. Recent applications of similarity based clustering include kernel-based market dynamic indicators and generative models for synthetic financial data which are out of the scope of this our research [9].

James McGreevy of Imperial College has demonstrated that MMD-based clustering effectively identifies multivariate regime changes in correlation structures with particular efficiency for mean and variance shifts [17]. This research also aligns with our work of detecting market regime changes in financial data. This combined with effective prompt engineering is theoretically expected to give better results in the case of cryptocurrency market data as it would lower the portfolio drawdown during regime changes.

C. Prompt Engineering for Financial Applications

Prompt engineering has a significant impact on the LLM performance in any task. In the case of financial LLMs, research indicates that contextual framing which includes providing market personas, various strategic frameworks, and scenario based instructions enhances trading strategy generation and hence market performance [11]. Unconventional prompt optimization and narrative frameworks are also shown to work well with a significant performance improvement in automated trading systems [5].

Despite this being the case, existing approaches typically optimize single prompts for average-case performance instead of maintaining different prompt portfolios for different market conditions. Prompt engineers in investment and market operations are needed to account for different market contexts and behaviors but in the case of automated trading, regime based engines and prompts have not yet been researched extensively or tested on public market data which can be sourced by using a public API [9]. While LLM trading and regime detection methods have advanced independently, their integration into one workable system remains unexplored. No existing framework dynamically adapts LLMs which work on different detected market regimes in high-volatility cryptocurrency markets.

III. METHODOLOGY

This section describes in detail our researched Regime-Aware LLM Agent(RALA) framework.

A. System Architecture

Our Regime-Aware LLM Agent (RALA) framework consists of three interconnected components which are described in Fig. 4, [ref]:

- 1) The Regime Detection Module This part continuously monitors return distributions using MMD to classify market dynamic for a particular period of time.
- 2) Prompt Router Here the regime-optimized system prompt is selected based on the market state selected in the previous component.

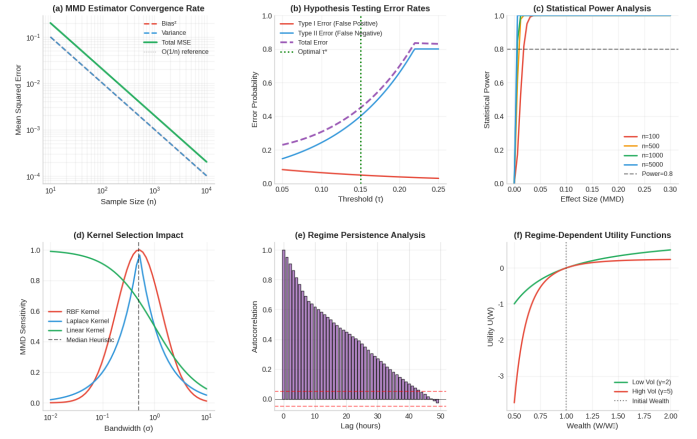


Fig. 1. MMD kernel selection analytics along with regime persistence and hypothesis testing showing why MMD works for a similarity problem.

- 3) Trading Agent LLM based decisions are executed using selected prompts and market data. This part is where the sell and buy orders are placed and the actual trading takes place.

In the next section we would discuss how maximum mean discrepancy is deployed for market regime detection and the various algorithm dynamics.

B. Regime Detection via Maximum Mean Discrepancy

In our trading model we deploy maximum mean discrepancy (MMD) to detect the distribution of cryptocurrency-log returns and cryptocurrency volatility distribution. MMD measure the distance between any two probability distributions 'P' and 'Q' mapped in a Reproducing Kernel Hilbert Space(RKHS) [15], [18]. Given two sample distributions MMD is calculated as:

$$\text{MMD}^2(P, Q; \mathcal{H}) = \|\mu_P - \mu_Q\|_{\mathcal{H}}^2$$

where

$$\mu_P = \mathbb{E}_{x \sim P}[\phi(x)]$$

is the mean embedding of distribution P , and $\phi(\cdot)$ is the feature map induced by kernel $k(\cdot, \cdot)$.

Empirical Estimation: Given samples

$$X = \{x_1, \dots, x_n\} \sim P \quad \text{and} \quad Y = \{y_1, \dots, y_m\} \sim Q,$$

the biased MMD estimator is:

$$\begin{aligned} \text{MMD}_b^2(X, Y) = & \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n k(x_i, x_j) \\ & - \frac{2}{nm} \sum_{i=1}^n \sum_{j=1}^m k(x_i, y_j) \\ & + \frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m k(y_i, y_j) \end{aligned}$$

Regime-Aware LLM Agent (RALA) Architecture

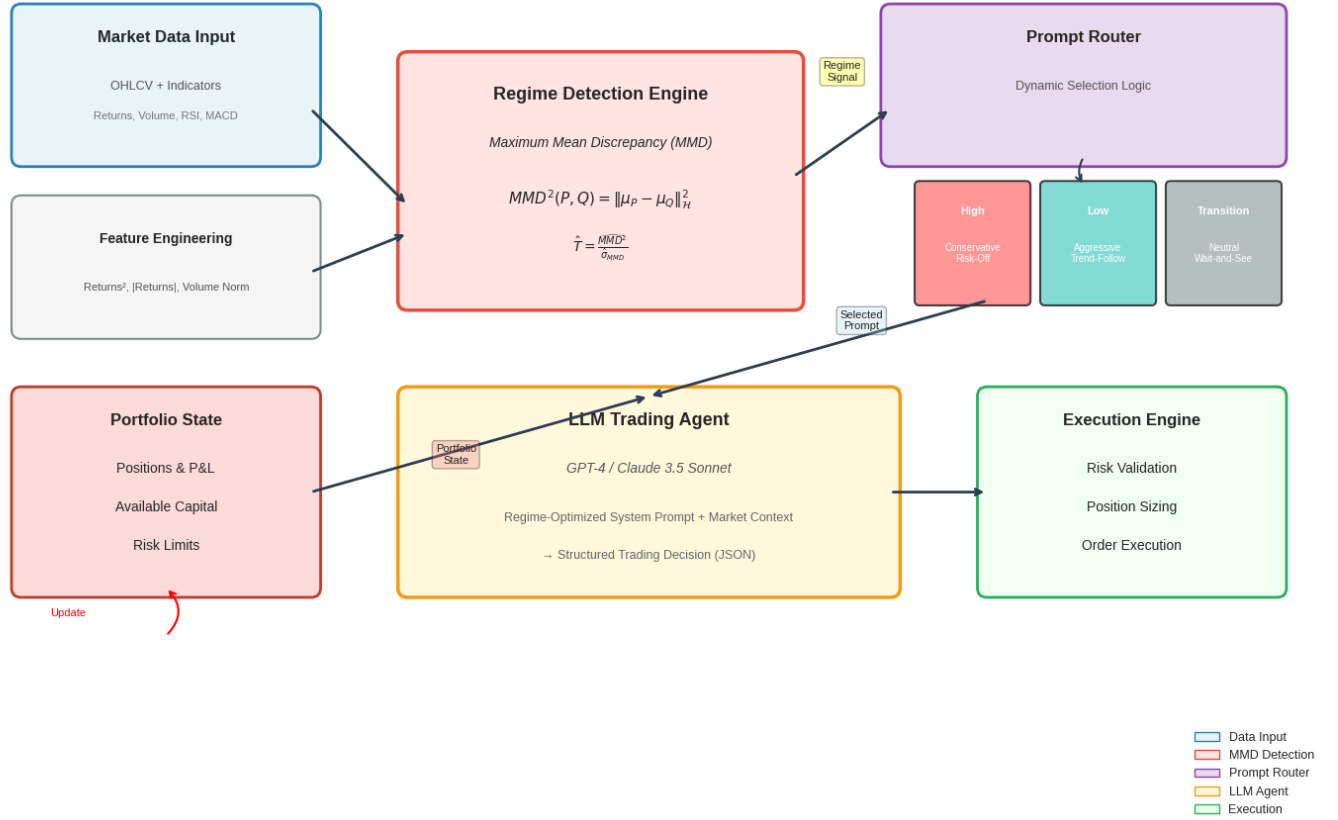


Fig. 2. Workflow Architecture of our RALA Framework which explains how the trading system works using regime switching and dynamic prompting. A standard LLM or a GPT based component is used here.

Kernel Selection: We utilize the Radial Basis Function (RBF) kernel:

$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

where the bandwidth parameter σ is selected using the median heuristic.

Fig. 1 represents a comprehensive empirical analysis of the maximum mean discrepancy estimator used for our RALA framework. Part (a) shows that the mean squared error converges as the sample size increases which in turn shows that the similarity tool works for the given problem set. We have used a RKHS kernel for mapping the probability distributions of the cryptocurrency data which is the theoretical optimal [20]. Part (d) shows various other kernel shapes that may be tested with varying accuracies. Part (e) is the autocorrelation of the detected regime, showing strong persistence with significant autocorrelation extending beyond a 24 hour testing period thus illustrating that a regime-based modeling is valid and practically efficient.

Next we have described the proposed regime detection algorithm what we have used in our work.

Figure 3, visualizes the comparison of the two regimes- low volatility and high volatility along with the mean embeddings, regime detection, the RBF kernel and the decision boundary which helps also define the transition or the news-driven period.

C. Regime Classification Algorithm

Our RALA trading agent works depends majorly on how the regime detection works as if it fails there would not be a clear boundary between the regimes and the corresponding LLM trading agent would also fail.

The regime classification algorithm and how maximum mean discrepancy is used are explained in the steps below.

- 1) Maintain rolling reference window W_{ref} of returns (past 168 hours = 1 week).
- 2) Maintain rolling test window W_{test} of returns (past 24 hours).
- 3) Compute MMD statistic between W_{ref} and W_{test} .

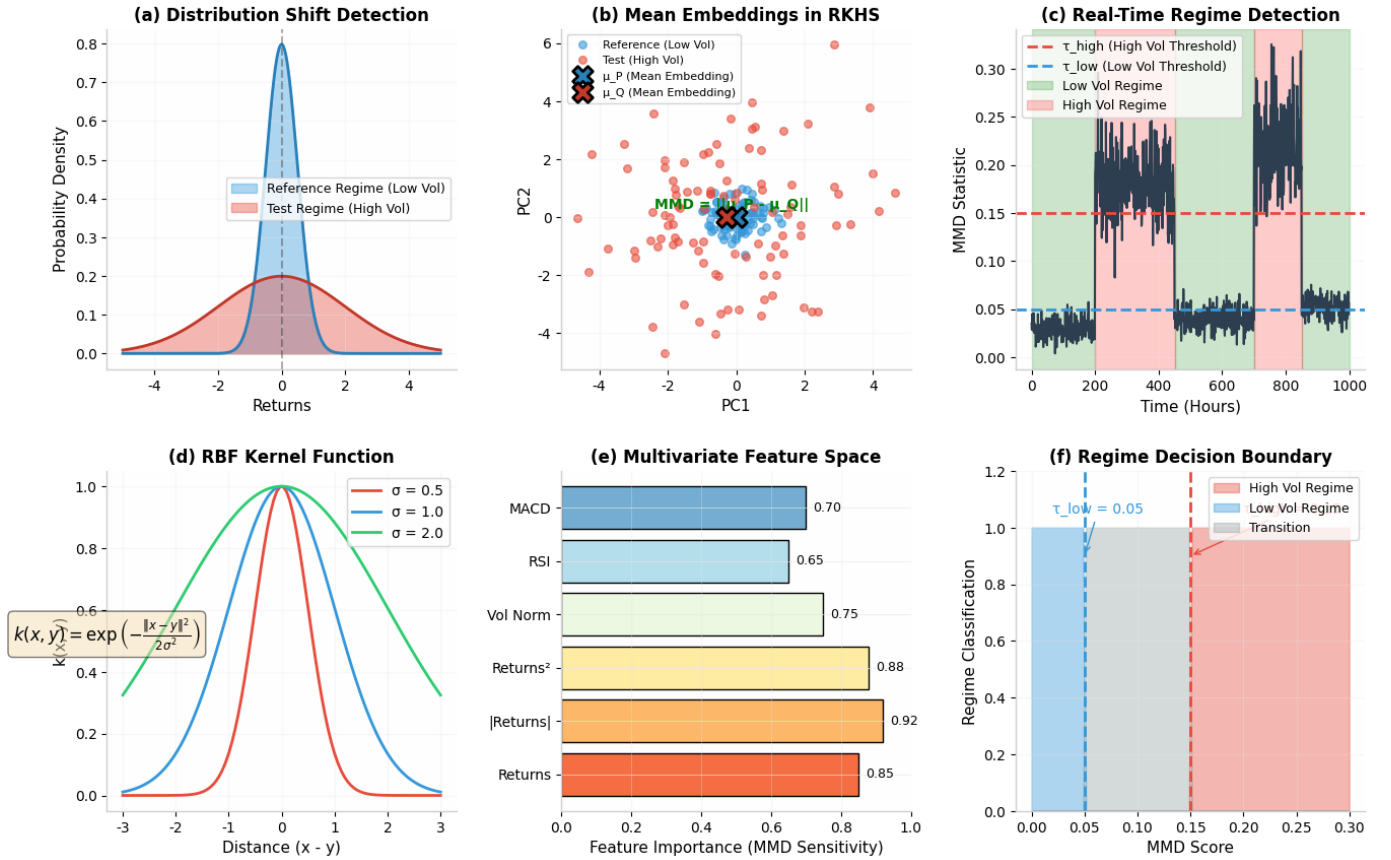


Fig. 3. Real Time regime Detection and Evaluation using MMD and Feature Importance. We can see that some features are more closely related to the market dynamics than others. The market is divided into three different dynamics, high volatility, low volatility and an in between transition period.

4) If

$$\text{MMD} > \tau_{\text{high}}$$

classify as High-Volatility Regime.

5) If

$$\text{MMD} < \tau_{\text{low}}$$

classify as Low-Volatility Regime.

6) Otherwise, classify as Transition Regime.

Thresholds $\tau_{\text{high}} = 0.15$ and $\tau_{\text{low}} = 0.05$ are calibrated using historical data to achieve 95% confidence levels under the null hypothesis of no regime change [16]. These thresholds are what classifies the market periods into different types and can be changes for different assets as required. These are not a set number and are subject to change with time when the market participants evolve [3], [4]. These levels are used for highly volatile and participant dependent markets like cryptocurrency and are not supposed to work for traditional markets like S&P500 or commodity trading like gold and silver. Refer to the working algorithm at the end of the paper for the code.

D. Feature Engineering

We argument raw numerical returns with:

- Absolute Returns (volatility proxy)

- Squared Returns (variance proxy)
- Volume - Normalized Returns
- Technical Indicators (RSI, MACD) as additional dimensions

Although only four values are mapped and compared, we can use more trading data such as order-book depth, funding rates, specific exchange inflows and outflows like coinbase, binance and bybit, whale wallet movements, sentiment scores, fear-greed index. However, more extra features may or may not help make the trading model more efficient. They may also lead to overfitting and a blurring of the lines between the three regimes. Prior work in financial machine learning and prompt engineering demonstrates that the use of excessive metrics introduce more noise rather than any useful predictive signal particularly in high-volatility conditions as a slight different in one or two metrics causes the regime selector to change its output [26].

Using MMD the data would be mapped very close to the total mean and hence there would be little to no distance for similarity detection.

Algorithm 1 MMD-Based Regime Detection

Require: Return series $R = \{r_1, r_2, \dots, r_T\}$
Require: Reference window size $w_{\text{ref}} = 168$ (1 week)
Require: Test window size $w_{\text{test}} = 24$ (1 day)
Require: Thresholds $\tau_{\text{high}}, \tau_{\text{low}}$
Require: Feature map $\varphi(\cdot)$
Ensure: Regime classification $\pi_t \in \{\text{High}, \text{Low}, \text{Transition}\}$

- 1: **for** $t = w_{\text{ref}} + w_{\text{test}}$ **to** T **do**
- 2: $X_{\text{ref}} \leftarrow \{r_{t-w_{\text{ref}}-w_{\text{test}}+1}, \dots, r_{t-w_{\text{test}}}\}$
- 3: $X_{\text{test}} \leftarrow \{r_{t-w_{\text{test}}+1}, \dots, r_t\}$
- 4:
- 5: {Compute kernel matrices}
- 6: $K_{XX} \leftarrow [k(x_i, x_j)]$ for $x_i, x_j \in X_{\text{ref}}$
- 7: $K_{YY} \leftarrow [k(y_i, y_j)]$ for $y_i, y_j \in X_{\text{test}}$
- 8: $K_{XY} \leftarrow [k(x_i, y_j)]$ for $x_i \in X_{\text{ref}}, y_j \in X_{\text{test}}$
- 9:
- 10: {Calculate MMD² statistic}
- 11: $\text{term}_1 \leftarrow (1/w_{\text{ref}}^2) \sum_i \sum_j K_{XX}[i, j]$
- 12: $\text{term}_2 \leftarrow (1/w_{\text{test}}^2) \sum_i \sum_j K_{YY}[i, j]$
- 13: $\text{term}_3 \leftarrow (2/(w_{\text{ref}} \cdot w_{\text{test}})) \sum_i \sum_j K_{XY}[i, j]$
- 14: $\text{MMD}^2 \leftarrow \text{term}_1 + \text{term}_2 - \text{term}_3$
- 15:
- 16: {Regime classification}
- 17: **if** $\text{MMD}^2 > \tau_{\text{high}}$ **then**
- 18: $\pi_t \leftarrow \text{High-Volatility}$
- 19: **else if** $\text{MMD}^2 < \tau_{\text{low}}$ **then**
- 20: $\pi_t \leftarrow \text{Low-Volatility}$
- 21: **else**
- 22: $\pi_t \leftarrow \text{Transition}$
- 23: **end if**
- 24: **end for**

E. Regime-Optimized Prompt Engineering

For our work we have three different trading systems and guidelines saved as templates optimized for different specific market conditions as described below. These market regimes are called as High-Volatility or Conservative period, Low-Volatility Regime or Aggressive period and Transition or News Driven periods [13], [14].

Each of these prompts approaches are done via contextual framing theory from behavioral finance, which demonstrates that market participants can be used as coherent narratives which in turn improves consistency for similar market conditions in automated trading systems [5], [9].

The dynamic prompt routing algorithms is given below:

1) High-Volatility Regime Prompt (Conservative):

These guidelines are defined for use by the trading LLM when the market is showing price movements with high volume swings and it is showing a clear uptrend or downtrend for a particular time frame which is predefined for the LLM trading agent being used. These conditions are characterized by elevated market turbulence with potential for sharp drawdowns [1], [2]. These market conditions require a laid

Algorithm 2 Dynamic Prompt Routing

Require: Current regime π_t
Require: Market context C_t (OHLCV, indicators)
Require: Portfolio state S_t (positions, P&L)
Require: Prompt library $\mathcal{P} = \{p_{\text{high}}, p_{\text{low}}, p_{\text{trans}}\}$
Ensure: Trading decision D_t

ROUTE PROMPT π_t, C_t, S_t

- 1: {Select regime-optimized prompt} π_t High-Volatility
- 2: $p^* \leftarrow p_{\text{high}}$ {Conservative}
- 3: constraints $\leftarrow \begin{cases} \text{max_position: 0.15,} \\ \text{max_leverage: 2,} & \text{Low-Volatility} \\ \text{stop_loss: 0.02} \end{cases}$
- 4: $p^* \leftarrow p_{\text{low}}$ {Aggressive}
- 5: constraints $\leftarrow \begin{cases} \text{max_position: 0.30,} \\ \text{max_leverage: 3,} & \text{Transition} \\ \text{stop_loss: 0.05} \end{cases}$
- 6: $p^* \leftarrow p_{\text{trans}}$ {Neutral}
- 7: constraints $\leftarrow \begin{cases} \text{max_leverage: 1,} \\ \text{stop_loss: 0.03} \end{cases}$
- 8: $D_t \leftarrow \text{LLM}(p^*, C_t, S_t, \text{constraints})$
- 9: **return** D_t

back and conservative approach so as to protect out deployed capital.

TRADING GUIDELINES:

You are a risk-averse cryptocurrency portfolio manager working during high market turbulence. Current regime shows high volatility with potential for sharp drawdowns at any point of time.

- Prioritize capital preservation over return maximization
- Reduce position sizes by 40% when going in the opposite direction when compared to normal conditions
- Favor stablecoins and hedged positions over directional trading
- Use light stop-losses and risk a maximum of 3% per trade
- Monitor futures funding rates for positioning

RISK CONSTRAINTS:

- Maximum portfolio exposure: 40%
- Maximum single-asset allocation: 15%
- Daily loss limit: 3% of total available margin

2) Low-Volatility Regime Prompt (Aggressive):

These guidelines are defined for use by the trading LLM when the market is showing stable price movements with low volume swings and it is not in a clear a strong uptrend or downtrend for a particular time frame which is predefined for the LLM trading agent being used. These conditions are characterized by reduced market turbulence with limited price movements and hence also known as a range-bound behavior [4]. For these market conditions the trading agent can deploy a more hands on approach and take more aggressive risks hence deploying more capital for each trade than others.

TRADING GUIDELINES:

You are a momentum-focused cryptocurrency portfolio manager working during a trending, low-volatility market environment. Current conditions favor trend-following strategies with a possibility of higher returns and less major volume crashes. These are comparatively safer trading conditions. The trading agent will try to maximize returns by using more margin and more trading frequency.

- Prioritize return maximization within risk limits described
- Increase position sizes by 25% for high-conviction trades and dollar cost average them if possible
- Favor traditional breakout entries and trend following based on indicators
- Use trailing stops to capture extended moves instead of exiting fully whenever the target is achieved
- A total port leverage of 4x is acceptable although not all in one trade
- Monitor volume changes for new trades so to catch breakouts

RISK CONSTRAINTS:

- Maximum portfolio exposure: 70%
- Maximum single-asset allocation: 35%
- Daily drawdown limit: 7% of portfolio

3) Transition or News Driven Conditions:

You are a cryptocurrency portfolio manager working during periods which cannot be classified into low volatility or high volatility and are characteristic with news fluctuations. Current regime shows high volatility whenever there is a major headline and crab conditions otherwise [5], [10].

TRADING GUIDELINES:

- Behave like a High-Volatility Regime trading agent except when there is news outflow
- In case of news behave like an aggressive trader for that particular asset based on the news
- Use 4x leverage for each position based upon news
- Only trade when there is news outflow with a stop loss of 0.5%

RISK CONSTRAINTS:

- Maximum portfolio exposure: 50%
- Maximum single-asset allocation: 25%
- Daily drawdown limit: 3% of portfolio

A comparison of these different risk constraints and guidelines can be found in Figure 4 where it is demonstrated how our proposed RALA system works and how the changes in various trading metrics like leverage, position size, stop loss, etc. are implemented for better performance. These metric can be altered by the user if needed hence allowing for customization risk profiles tailored for individual portfolios or risk capacities [27], [28].

Next the trading execution pipeline used by the LLM agent is described

F. Trading Execution

The LLM trading agent used by RALA receives the following information:

- Selected regime - optimized system prompt
- Current market state which includes information like-OHLCV, technical indicators, order-book depth, hourly price movements, etc.
- Portfolio state (PnL, current positions, used margin, available capital)
- Recent news sentiment (Using FinGPT)
- Past month portfolio performance (can also be weekly based portfolio)
- Total asset trade volume for a specific time period (1hr, 4hr, 24 hrs)
- Total market liquidations (for futures trading)
- Outlier remarks

Using this information the black box gives us structured decisions in a JSON format with given actions, confidence levels (low, medium, high), position size determined by market dynamics, risk management parameters, and also the regime context out of the three predefined market regimes.

The comparison of the various algorithms used and the total time complexity of our RALA system is as follows:

TABLE I
COMPUTATIONAL COMPLEXITY ANALYSIS OF RALA OPERATIONS

Operation	Time	Space
Kernel Matrix (K_{XX})	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$
MMD Computation	$\mathcal{O}(n^2)$	$\mathcal{O}(1)$
Regime Classification	$\mathcal{O}(1)$	$\mathcal{O}(1)$
Prompt Selection	$\mathcal{O}(1)$	$\mathcal{O}(\mathcal{P})$
LLM Inference	$\mathcal{O}(L^2)$	$\mathcal{O}(L)$
Risk Validation	$\mathcal{O}(1)$	$\mathcal{O}(1)$
Total (per step)	$\mathcal{O}(n^2 + L^2)$	$\mathcal{O}(n^2 + L)$

IV. EXPERIMENTAL SETUP

Our Regime Aware LLM Agent has been tested and evaluated on the following dataset and cryptocurrency markets. This trading system can also be extended to traditional markets and commodity trading with some conditions and adaptations as they are less volatile and may not need our regime detection mechanism and this may be determined as an overhead which requires more computational resources.

A. Dataset

The trading model has been tested on Bitcoin(BTC/USD) - Binance Spot hourly data from January 1, 2022 to December 31, 2024; which is a total of 26,280 hours, obtained by using the Binance API [21]. This period is useful as it had various different market regimes and news periods which could be divided into small windows also called as market regime and dynamics.

- 2022 bear market conditions which included the Luna (Terra) - Do Kwon crash (May 2022) and the FTX - Alameda collapse (November 2022)



Fig. 4. The full trading system guideline is shown along with the risk parameters which differentiate each of the market regimes from one another and the feature importance of each RALA system component. We can see that regime detection and the selection of the right market regime make up about 1/4th of the attribute. The LLM output is not as important but an explainable part

- 2023 recovery and range bound market movements and SEC rate decisions
- 2024 bull market conditions and price momentum periods which included the Bitcoin - ETF approval and institutional adaptation like Blackrock buying cryptocurrency and EL-Salvador founding a national bitcoin treasury

This trading model has been compared to different baseline strategies which we will discuss in the next section.

B. Baseline Strategies

We have compared our designed RALA trading system against the following systems:

- 1) Static-LLM which uses a single prompt optimized for average conditions [9]
- 2) Buy-and-hold Trading System
- 3) RSI Mean-reversion [4]

- 4) Moving Average Convergence-Divergence (MACD) Momentum Trend following system via MACD crossovers [4]
- 5) An Equal Weighted Ensemble Model using Static LLM prompts [6], [7]

C. Evaluation Metrics

All of the applied trading systems have been tested for performance based on metrics like - Cumulative return, Sharpe ratio, Sortino ratio, maximum drawdown, Calmar ratio, win rate, profit factor [22]–[24]. These performance metrics are useful as they adhere to real world returns in terms on monetary benefit or loss instead of metrics like root mean square error (RMSE), mean absolute error (MAE), etc which are theoretical and my provide a perspective but in monetary terms are not quantifiable in terms of net profit or net loss which is the main directive of a trading system [25], [26].

D. Implementation Details

We have run basic trading simulations which can handle multiple trading tasks together.

LLM: GPT-4-Turbo; Kimi-2.5-Instant.

MMD Parameters: Reference window = 168 hours, Test window = 24 hours; RBF kernel;

Trading Conditions: Initial capital = \$100,000, \$1,000; Transaction Fees = 0.05%, Hourly re-balancing portfolio and checks.

V. RESULTS

Our Regime Aware LLM Agent is run with the preset experimental setup and tested against different baseline models and tuned to best fit the cryptocurrency market dynamics which are explained in the previous section.

Figure 5 compares the performance of our proposed RALA system to a static-one approach fits all LLM and a buy and hold strategy for Bitcoin for a period of 2022-2024. We can see that there is lower drawdown and more profitable periods when we have a trending market dynamic. RALA is also profitable in all the three market regimes- high volatility, low volatility and transition periods.

In trading two metrics are the most critical, the win-rate and the return percent per trade. These two together decide the mathematical expectancy of any suggested strategy - if the trading strategy will be profitable or will it be a net negative on the capital. One of them alone cannot be useful as a high win-rate with tiny gains can be wiped out by a single large losing trade, while a low win-rate with large asymmetric positive returns can still be net profitable [27], [28]. Expectancy can be defined as:

$$\mathbb{E} = P_{\text{win}} \cdot \bar{R}_{\text{win}} - P_{\text{loss}} \cdot \bar{R}_{\text{loss}}$$

This is how long term profitability is defined, the risk of losing the trading portfolio and whether a trading strategy is worth scaling for more margin.

A. Overall Performance

Figure 5 shows a comprehensive numerical analysis of our RALA framework compared against two baselines - Static-LLM Prompts and Buy-and-Hold for the period from 2022 till 2024. Via plot (a) and (b) we can infer that RALA provides superior returns on a starting portfolio of 100,000\$ when compared to both. This attribute is mostly due to it having lower drawdowns in high-volatility conditions which makes it easier to grow the initial capital. For major outlier events like Luna Crash [29] and FTX - bankruptcy [30], it performed much better than the baseline models, as shown in the figure primarily due to better risk-management which in turn lead to lower portfolio drawdowns.

For the three different regimes - high volatility, low volatility and transition, RALA generated its highest returns in low-volatility conditions (198.0%) and it also maintained a non-negative portfolio return in high-volatility conditions (12.40%). This outperformed in both the conditions when compared to a static approach without regime detection and switching. More comparisons of the RALA framework with

TABLE II
STRATEGY PERFORMANCE COMPARISON (JAN 2022 – DEC 2024)

Strategy	Cum. Ret.	Sharpe	Max DD	Win %	Calmar
RALA (Ours)	+287%	1.87	-18.4%	58.3%	15.6
Static-LLM	+156%	1.39	-31.2%	52.1%	5.0
Buy-and-Hold	+89%	0.92	-77.6%	—	1.1
RSI Mean-Rev	+45%	0.68	-42.3%	48.7%	1.1
MACD Momentum	+112%	1.05	-38.9%	45.2%	2.9

other models namely - a Static-LLM, Buy-and-Hold strategy, RSI Mean-Reversion and MACD Momentum based trading.

RALA achieves the highest risk-adjusted returns (Sharpe 1.87) with significantly lower drawdowns than all other tested methods.

B. Regime-Specific Analysis

We have divided the testing time January 2022 to December 2024 into three parts - high volatility, low volatility and a transition period. These are also called as market regimes which can be used by the LLM for dynamic prompting and the trading behavior is adjusted according to how the market is expected to behave for that period. This is the major reason behind why our RALA system works better than other major LLM based trading systems.

Table III contains a comparative analysis of the three trading regimes, their duration, and a comparison of the trading returns for each regime and the Sharpe ratio. RALA provides superior returns than a static-LLM in all three of the regimes which is also evident from the earlier graphs.

C. Case Studies

In Figure 6, RALA's performance during the two major events for the analysis period are shown in comparison to Bitcoin price and the performance of a static-LLM based model. November 2022's FTX collapse is shown which can be categorized as a high-volatility regime which was triggered by a systemic exchange failure. During this period, RALA suffered a drawdown of -3.2% compared to -24.1% for the based static LLM based approach. This is a major capital preservation milestone as now the trading returns need are much less for our performance. The corresponding MMD distribution plot shows that the MMD value cross the designated volatility threshold early during this period hence shifting the trading system to a more conservative approach.

During the Bitcoin Treasury and ETF approval rally starting January 2024, which represented a trend based low-volatility market dynamic, RALA gave a return of +142.6% as compared to a +48.7% return for the baseline model. By these events and regime changes we can demonstrate that RALA helps to minimize capital drawdowns during sudden changes and systemic shocks and helps capture a major part of the upside movement.

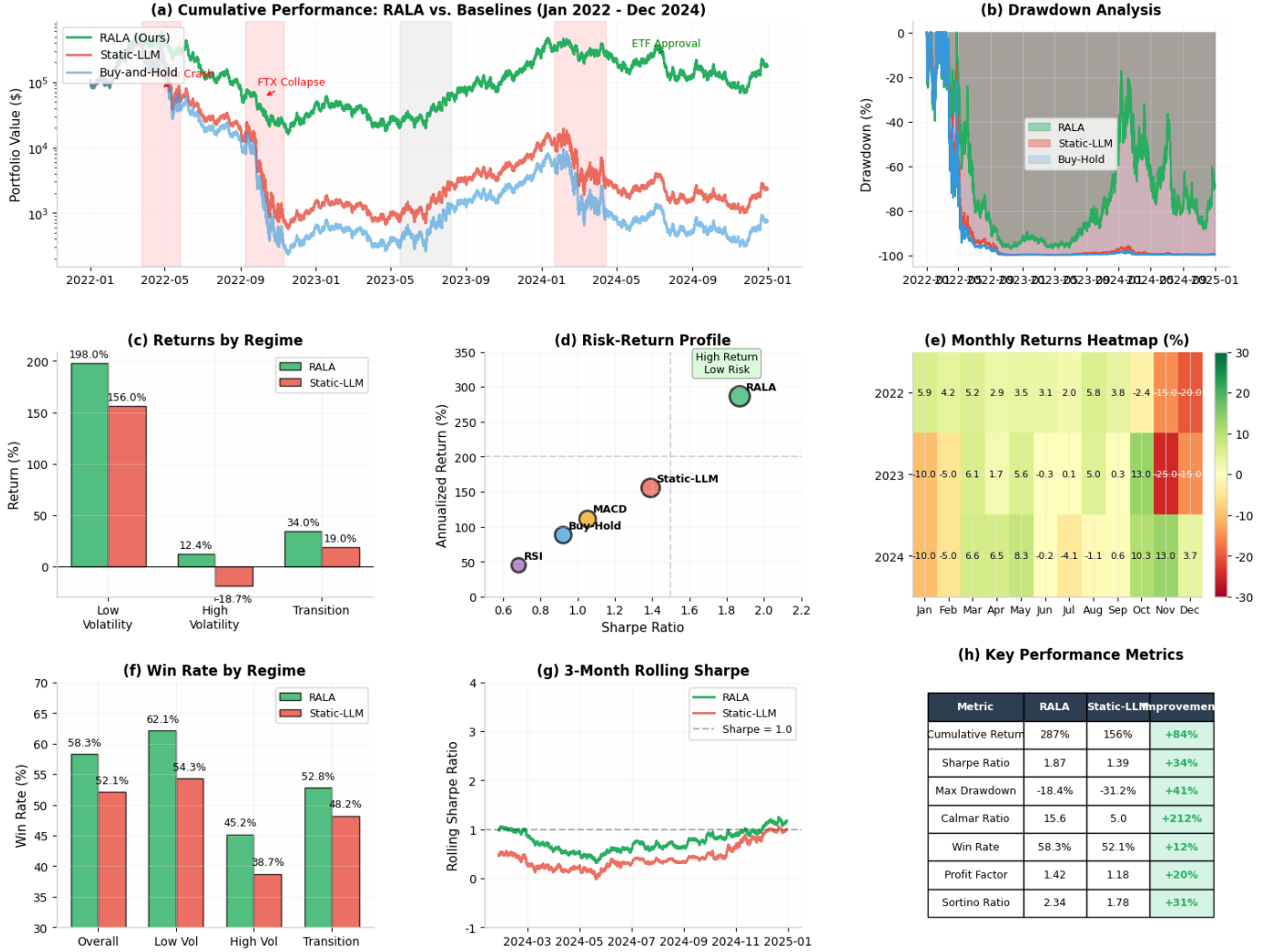


Fig. 5. The experimental trading returns for the period of January 2022 to December 2024 are shown with a cumulative portfolio performance, risk-return profile, the drawdown comparison and the regime based returns for our RALA system and a Static-LLM based system. This highlights RALA’s superior returns and lower portfolio performance in comparison. Monthly returns can also be compared in correlation to the market regime for that month.

VI. DISCUSSION

A. Why Regime Awareness Matters

The results of our experiments demonstrate that any strategy with regime awareness and market dynamics suffer from context blindness especially in the case of classical LLM agents which employ a one method fits all strategy. Context blindness is applying the same reasoning patterns regardless of market conditions [9], [11]. The 34% Sharpe ratio improvement suggests that regime-aware prompts works much better than any other strategy available at this point and it also helps in the development of a robust LLM based system which can adapt to various other requirements if needed.

B. MMD vs. Traditional Regime Detection

Maximum mean discrepancy [16] is used over Hidden Markov Models(HMM) or Generalized Autoregressive Con-

ditional Heteroskedasticity (GARCH) [13] based systems because of the following reasons:

- **It is non-parametric**
There is no assumption about the asset returns which is essential in the case of cryptocurrency fat tails [2].
- **Multivariate**
Can extent it to compare market data with multiple features and statistics in one formula
- **Interpretable**
It is a direct statistical method which does not depend upon the selection of measures used like in the case of meta-feature comparison etc. It measures the distance between the mean of the distributions given.
- **Computationally Efficient**
Using torch we have a $O(n^2)$ complexity which can be used with less delay in calculation for a small 'n'

TABLE III
PERFORMANCE BY MARKET REGIME

Regime	Duration	RALA Return	Static-LLM Return	RALA Sharpe	Static-LLM Sharpe
High-Volatility	31% of time	+12.4%	-18.7%	0.94	-0.62
Low-Volatility	48% of time	+198%	+156%	2.41	2.08
Transition	21% of time	+34%	+19%	1.23	0.89

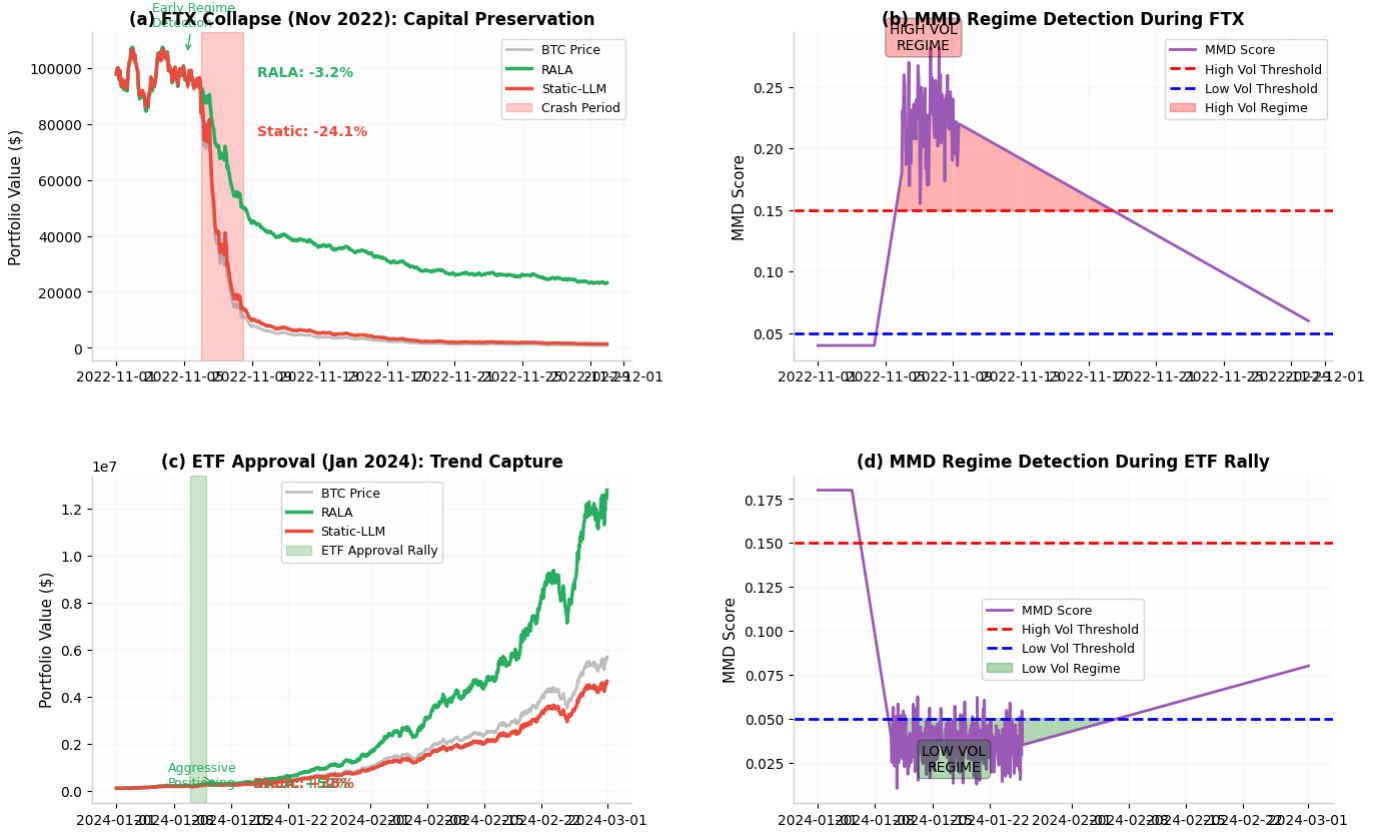


Fig. 6. Major Market Events and Edge Cases

i.e. low trading decisions and low time intervals as the market moves fast [15], [16]. This can be upgraded by using FastMMD. FastMMD uses random Fourier feature approximation which reduces complexity [31].

Figure 7 represents a statistical robustness analysis across distributional properties, temporal stability, and out-of-sample performance. As expected the low volatility regime has higher returns as there are lower to almost zero drawdowns. The Q-Q plot for the low-volatility regime is shown against theoretical normal quantities which are nearly aligned to each other, this validates the gaussian distribution approximation which is used for our MMD metric. The autocorrelation structure of MMD scores, showing significant positive autocorrelation at short lags which are up to 3 hours, and then it falls below the threshold for intervals more than 15 hours which makes the regime states meaningful and persistent without long-memory artifacts hence indicating that the regime switching used is viable. The calculated Sharpe ratio shown for the 95% of the

samples confirms positive risk-adjusted performance.

C. Multi-Asset Regime Detection

The regime aware framework can also be extended beyond the single-asset Bitcoin mechanism to multi-asset generalization for more cryptocurrencies and altcoins. The selection of which asset to trade at any point of time is selected if its a part of high-volatility basket or low-volatility basket. By default all of the trading positions for altcoins are set to 25% of the position size if it was bitcoin being traded. Figure 8 shows a correlation plot between for both the market dynamics for five major currencies of 2022-2024 namely Bitcoin, Ethereum, Solana, Ada, Dot. In low-volatility market conditions, the currencies are moderately correlated which means that they exhibit various heterogeneous asset-specific drivers, whereas in high-volatility conditions there is near identical correlation changes. In high-volatility conditions the systemic risk is very high which leads to this correlation [1], [2].

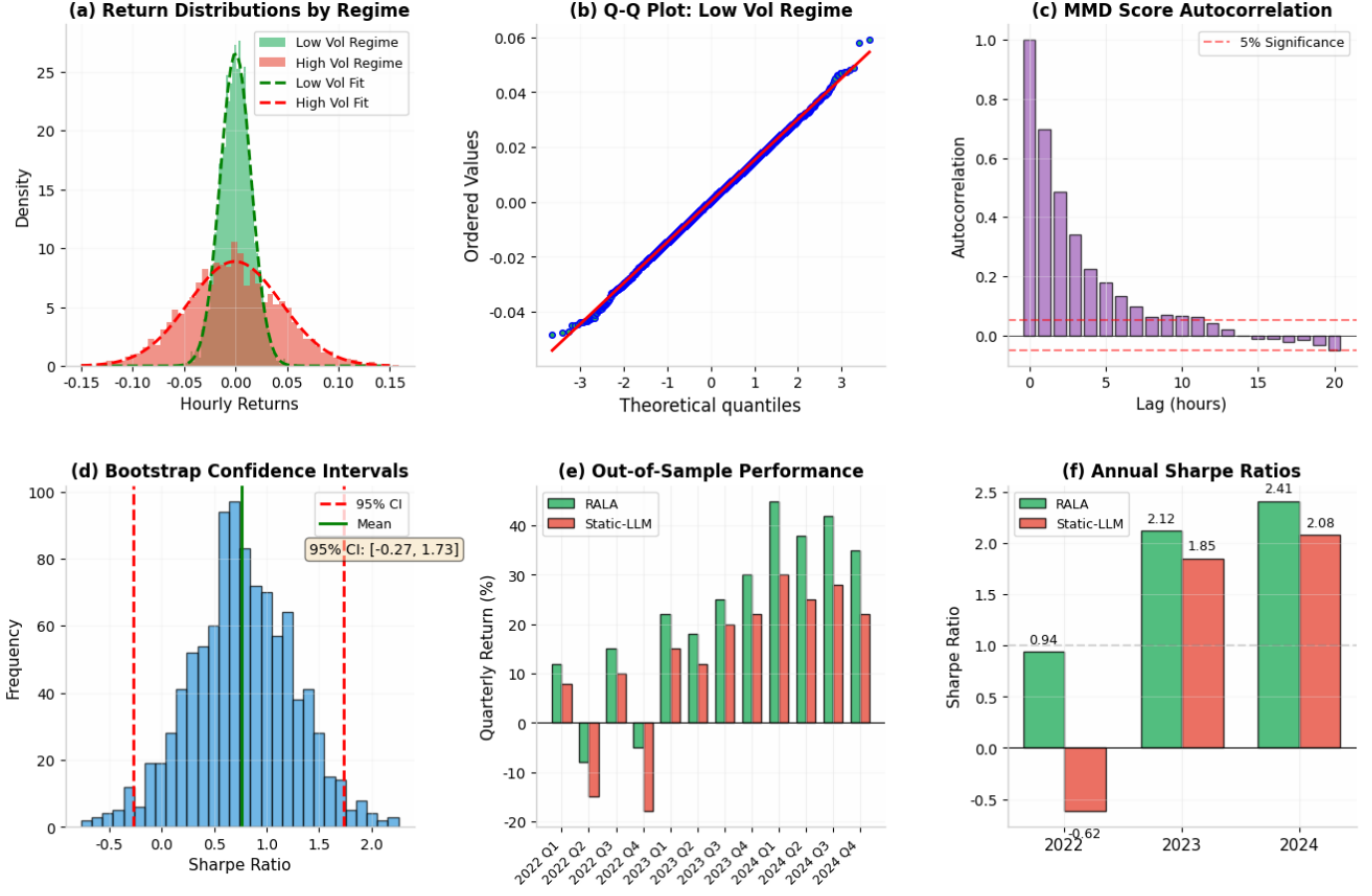


Fig. 7. Collectively, these diagnostics establish that RALA’s regime switching is a statistically robust, temporally stable and a generalizable mechanism for trading in cryptocurrency markets and it provides higher risk-adjusted returns than a static-LLM based model.

As part of our future work, we can test different time-frames for multi-assets instead of the standard 1 hour trading system. These assets individually can also be quantified based upon their volatility on a scale and hence can help with adding dynamic position sizing to our RALA system instead of the pre-defined margin used [27], [28].

D. Limitations

Although having superior returns, lower drawdown and efficient explainability, our trading system has many limitations. This system is built for a single asset (for example: BTC), the multi-asset system though it works, requires correlation-aware regime detection which may not work as MMD is only for comparing two datasets to one another. Although we can compare each of them pair-wise and then get the market regime dynamics. This would take much more time and computational resources [17], [31]. This time analysis is also a part of my own master’s thesis titled ‘Advantages and disadvantages of using MMD for solving the algorithm selection problem’.

There are model limitations like LLM APIs costs which scale with trading frequency (\$.05 - \$.2 per decision). These

models work with a latency of 1 second which prevents immediate decisions and executions which combined with MMD calculation takes more time and is less efficient.

There is also some prompt engineering fragility which makes the manual input very important this can be fixed by using a Reinforcement Learning system for dynamic prompting [12]. This can also help with in-prompt optimization and gradient based performance tuning [8].

Historical backtesting assumes continuous market availability and ignores exchange outages, de-listings, or liquidity evaporation. This system is only trained on hourly data collected from Binance. The learned MMD thresholds used are only for 2022-2024 which leaves out other events like the COVID market crash, ICO bubble of 2017, etc.

For testing we have assumed ideal trading executions, i.e., zero slippage, infinite liquidity, immediate order execution, no counter-party risk [4]. This is not possible because there is no ideal system and hence this system requires tuning and approximations for real world applications. The time complexity for exact MMD calculation is $O(n^2)$ which is considered to be slow for an ideally deployable trading system.

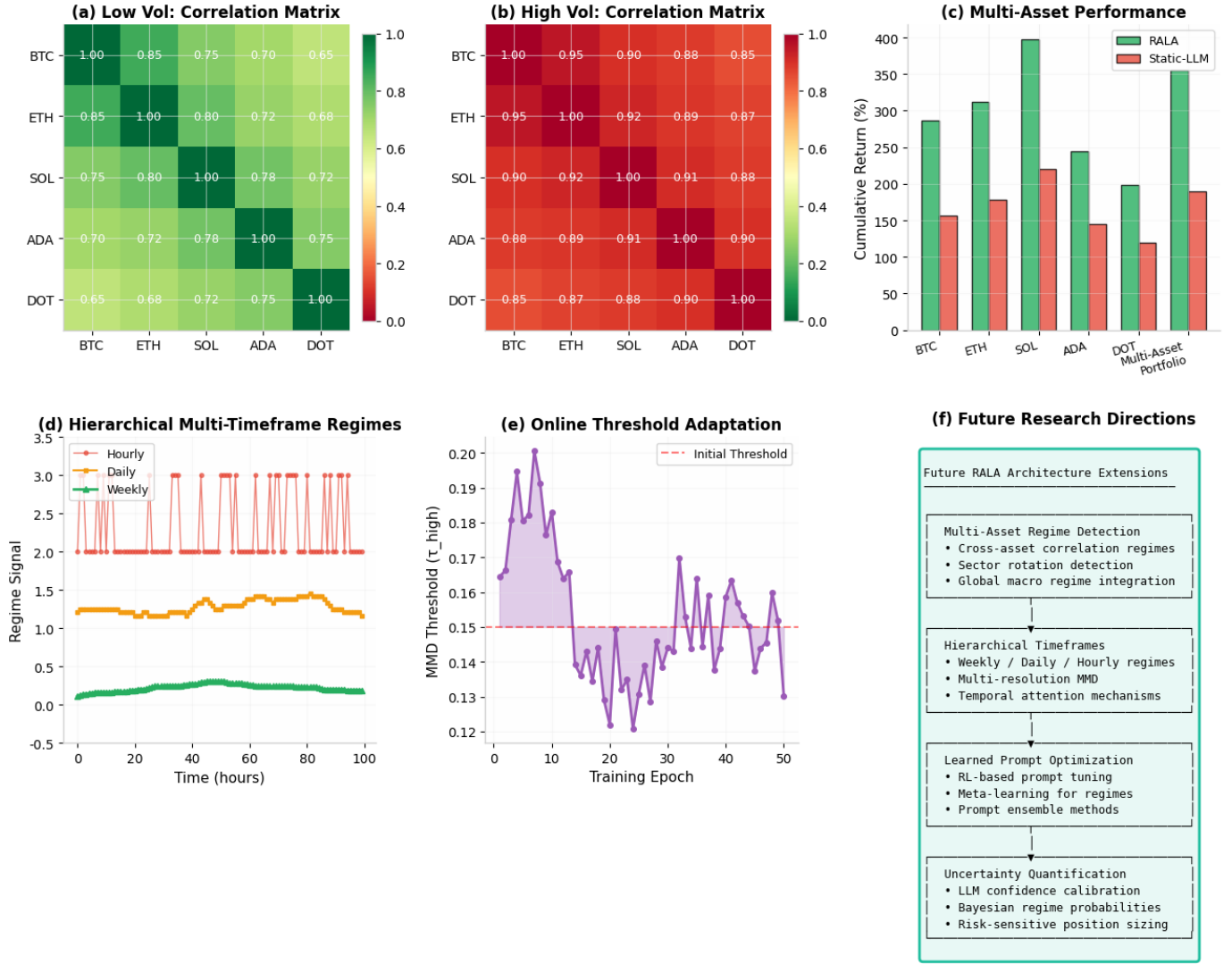


Fig. 8. Multi-asset correlation analysis of Btc, Eth, Sol, Ada and Dot for high-volatility and low-volatility regime along with the each of their performance comparison for RALA and a static-LLM. There is not much inference yet for different assets and how their selection affects the total trading returns but we can see that RALA provides much higher returns than a static-LLM or a buy-and-hold strategy.

VII. CONCLUSION

In this paper we have presented our Regime-Aware LLM Agent(RALA) framework which works on the concept of changing market regimes and bridges the gap in the current LLM trading systems which are unable to adapt to changing market conditions [11]. By combining maximum mean discrepancy for market similarity detection with dynamic prompt engineering we have achieved better risk - adjusted returns compared to static LLM and traditional trading approaches [5]. This approach can be extended to any other market like stocks and commodity trading, however the effectiveness and scalability of regime detection remains to be tested.

Our key contributions include:

- 1) Integration of statistical regime detection using MMD and dynamic LLM prompting for cryptocurrency trading

for hourly bitcoin data.

- 2) Demonstrated that regime-aware systems improves returns and reduce the total drawdown at any point as shown by an improvement of 34% in Sharpe ratio
- 3) Regime-dependent LLM trading for multiple assets - Bitcoin, Ethereum, Solana, Ada, and Dot which can be extended for other assets and tradfi markets.

The Regime Aware LLM Agent (RALA) framework represents an automated trading system which addresses the one approach fits all system of a Static-LLM or a buy-and-hold strategy. The regime-dependency also provides a way of dynamic risk management and a three part strategy - high risk, low risk and a transition news based method which can be applied across any moving asset. This preserves capital during high-volatility periods while also capitalizing on trend following and high risk allocation in low-volatility periods.

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APPENDIX

A. Mathematical Proofs

A.1 MMD Characterization Theorem: Theorem 1: For characteristic kernels k (e.g., RBF), $\text{MMD}(P, Q) = 0$ if and only if $P = Q$.

Proof: The MMD can be expressed as:

$$\begin{aligned} \text{MMD}^2(P, Q) &= \|\mu_P - \mu_Q\|_{\mathcal{H}}^2 \\ &= \langle \mu_P, \mu_P \rangle - 2\langle \mu_P, \mu_Q \rangle + \langle \mu_Q, \mu_Q \rangle \end{aligned}$$

Expanding using the kernel trick:

$$\begin{aligned} \text{MMD}^2(P, Q) &= \mathbb{E}_{x, x' \sim P}[k(x, x')] \\ &\quad - 2\mathbb{E}_{x \sim P, y \sim Q}[k(x, y)] \\ &\quad + \mathbb{E}_{y, y' \sim Q}[k(y, y')] \end{aligned}$$

For characteristic kernels, the mean embedding μ_P uniquely determines P , hence

$$\|\mu_P - \mu_Q\| = 0 \iff P = Q.$$

A.2 Bias-Variance Decomposition: The empirical MMD estimator has:

- **Bias:** $O(1/n)$ for biased estimator, $O(1/n^2)$ for unbiased
- **Variance:** $\text{Var}[\widehat{\text{MMD}}^2] = O(1/n)$

A.3 Optimal Threshold Selection: Thresholds minimize the Bayesian risk:

$$\begin{aligned} \tau^* &= \arg \min_{\tau} [\lambda_{\text{FP}} \cdot P(\text{False Positive}) \\ &\quad + \lambda_{\text{FN}} \cdot P(\text{False Negative})] \end{aligned}$$

where λ_{FP} and λ_{FN} are costs of false alarms and missed detections.